

GenAI - Code is on the last page in a table

ChatGPT - Free version, no think mode

Gemini - Free version using the 3.6 Flash version that was defaulted

It is important to consider that turning on modes for each GenAI may have resulted in different results as it makes the AI think longer thus coming up with different solutions,

Both were accessed on their respective website i.e <https://chatgpt.com/> and <https://gemini.google.com/app>

Prompt:

Write me a C program that picks a secret number between 1 and 10. Letting the user get up to 3 tries to guess the number.

For the requirements;

Use a fixed secret number

Use printf to prompt the user and scan to read their guess

Use a while or for loop to let the user guess up to 3 times

Use if-else statements to check if the guess is too high, too low, or correct.

Stop the loop immediately if the user guesses the correct number finally, print a final win or lose message

As an example for the output;

Guess a number between 1 and 10.

Attempt 1/3. Enter your guess: 5

Too low! Try again.

Attempt 2/3. Enter your guess: 8

Too high! Try again.

Attempt 3/3. Enter your guess: 7

Correct! You win!

Correctness

- When both codes were put into a file and compiled to run, they both worked exactly as intended and had the correct output.

execution time

- Gemini was slightly faster, more likely due to using a for-loop instead of a while loop to check how many times the user had guessed.

Space Complexity

- The space complexity is roughly the same although considering that we do have the final value for many guesses we have, instead of waiting just for a condition to be met it is far better to use a for loop that is most likely far easier to handle.

Maintainability

- Both codes were severely lacking in maintainability, neither of them had any comments of what was going on. To be fair, it can also be considered to be input for the number to be restricted between 1 and 10 since the program is directly telling the user this. However there is no way input like strings could be handled. In addition, I also liked how Gemini decided to use const before the two variables that needed to be constant, it adds extra emphasis on the fact that it shouldn't be modified.

My Fixes

Decision: I will decide to use Gemini for the source code.

Correctness

- Due to my lack of experience, I believe that the current code is alright, the output for gemini matched the expected output exactly in terms of conditions like being above or below. The only difference that doesn't make much of a difference is how it chooses to convey that text.

execution time

- Since the for loop is already being done, it would be redundant to change it to a while loop therefore this will also stay the same.

Space Complexity

- Similar to the execution time, there aren't any methods that I know of that could affect the space complexity in a major way from what it currently is.

Maintainability

- This is where most changes will occur, I hope to comment every single line of code in order to improve this aspect. Additionally I will also add another if statement so that the input doesn't take a string of letters from the user.

ChatGPT

```
#include <stdio.h>

int main() {
    int secretNumber = 7;
    int guess;
    int attempt = 1;
    int correct = 0;

    printf("Guess a number between 1 and 10.\n");

    while (attempt <= 3) {
        printf("Attempt %d/3. Enter your guess: ", attempt);
        scanf("%d", &guess);

        if (guess == secretNumber) {
            printf("Correct! You win!\n");
            correct = 1;
            break;
        } else if (guess < secretNumber) {
            printf("Too low! Try again.\n");
        } else {
            printf("Too high! Try again.\n");
        }

        attempt++;
    }

    if (!correct) {
        printf("Sorry! You lose. The secret number was %d.\n", secretNumber);
    }

    return 0;
}
```

Gemini

```
#include <stdio.h>

int main(void) {
    const int SECRET_NUMBER = 7;
    const int MAX_ATTEMPTS = 3;
    int guess = 0;
    int won = 0;

    printf("Guess a number between 1 and 10.\n");

    for (int attempt = 1; attempt <= MAX_ATTEMPTS; attempt++) {
        printf("Attempt %d/%d. Enter your guess: ", attempt, MAX_ATTEMPTS);
        scanf("%d", &guess);

        if (guess == SECRET_NUMBER) {
            printf("Correct! You win!\n");
            won = 1;
            break;
        } else if (guess < SECRET_NUMBER) {
            if (attempt < MAX_ATTEMPTS) {
                printf("Too low! Try again.\n");
            }
        } else {
            if (attempt < MAX_ATTEMPTS) {
                printf("Too high! Try again.\n");
            }
        }
    }

    if (!won) {
        printf("You ran out of tries! You lose. The secret number was %d.\n", SECRET_NUMBER);
    }

    return 0;
}
```